An Overview of Stepwise Project Planning

Reference: Hughes & Mike, Software Project Management,Tata McGrawhill.

Stepwise Project Planning

- **Step 1: Identify Project Scope and Objectives** Identify what is to be done and what is not required and also identify ways to measure the success in achieving objectives.
- Identify the person, or committee or board incharge of project.
- Identify stakeholders and their interests.
- Modify objectives as necessary based on stakeholder analysis
- Establish methods of communication with all stakeholders

Stepwise Project Planning contd..

Step 2: Identify Project Infrastructure

- Perform strategic planning for the project: Strategic Planning is the process of defining its strategy or direction and making decisions on the resources to be allocated

  Establish the framework within which the project/system is to be developed such as deciding hardware and software standards.

Stepwise Project Planning step 2 contd..

- Identify installation standards and procedures: Change control and configuration management standards, quality standards and procedures, project planning and control standards to be identified

  Documentation of procedures followed in developing the project
- Identify project team organization: identify who will be in which group, who will report to whom? Though the organizational structure is fixed, project leaders will have some control over it.

Stepwise Project Planning contd...

Step 3: Analyze Project Characteristics

- Is the project product based, service based or result based?
- Identify high level project risks: Most risks can be attributed to the operational or development environment, the technical nature of the project or the type of project to be created

  For example people are not interested to accept easily to new working environment or to new tools etc.

  Consider user requirements in implementation,
- Select appropriate development methodology (methods to be used) and life cycle approach.
- Review overall resource estimates (including effort and other resources to implement the project)

Stepwise Project Planning contd...

Step 4. Identify Project products and activities

4.1 Identify and describe project products: The products handed over to the client are called deliverables.

Other products are intermediate products and used in creating deliverables

Products could be software components, documents or a trained user.

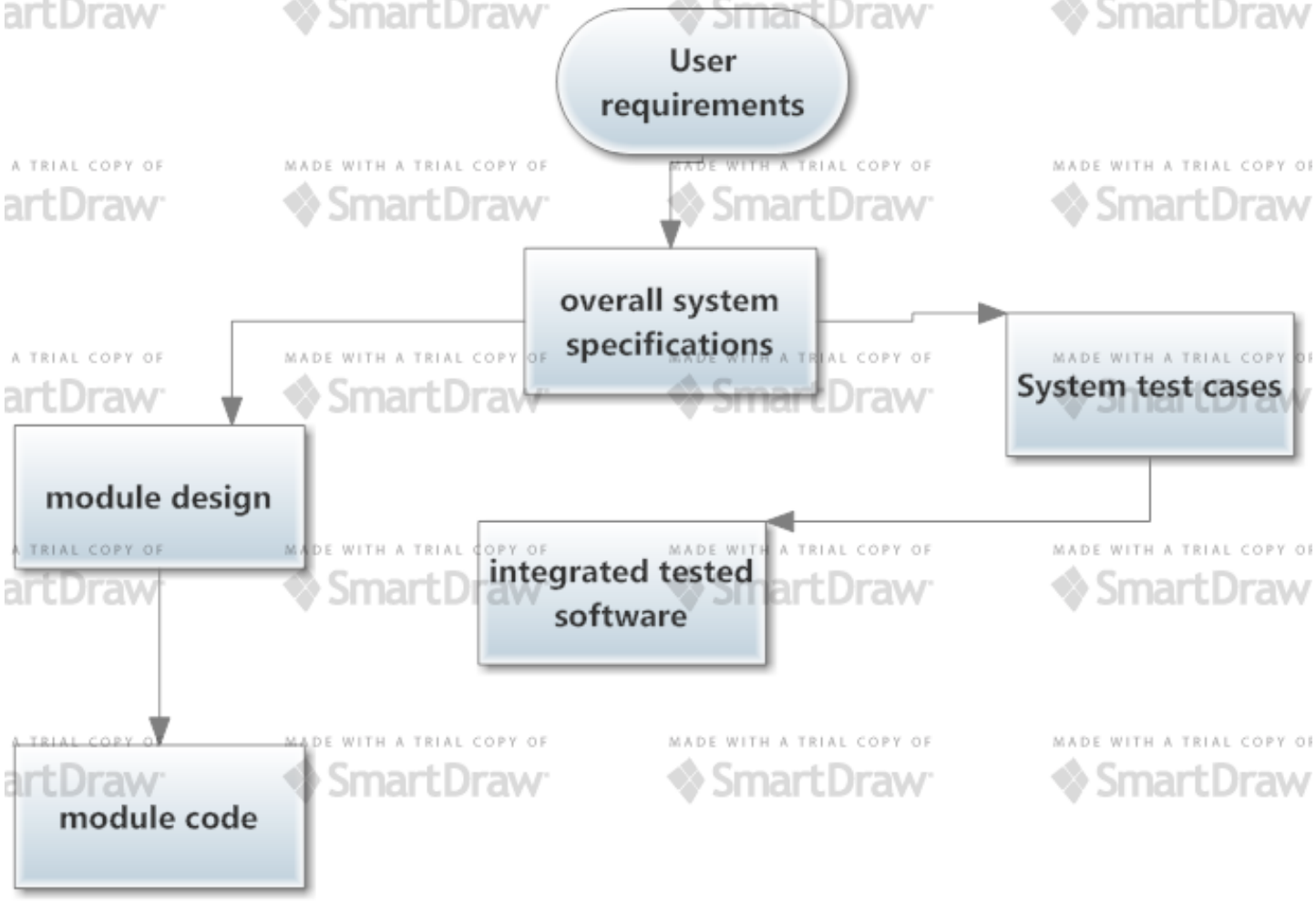
In Product breakdown structure, products could be grouped into those relating to system (system products) relating to individual modules (module products) and management products or progress reports.

Stepwise Project Planning Step 4 contd...

- System products: They include overall specification, test cases and tested software.
- Module products: They include module design and module code
- Management products: These could be progress reports dynamically created
- **4.2 Document generic product flows:** Some products need other products to exist before they can be created. For example system design is needed before programs can be written.

  Program specification must exist before the design. The relationship can be shown in product Flow Diagram (PFD) as shown in the next slide

Example of a Product Flow Diagram



Step 4 contd..

4.3 Recognize product Instances: More than one instance of a product type will have same PFD

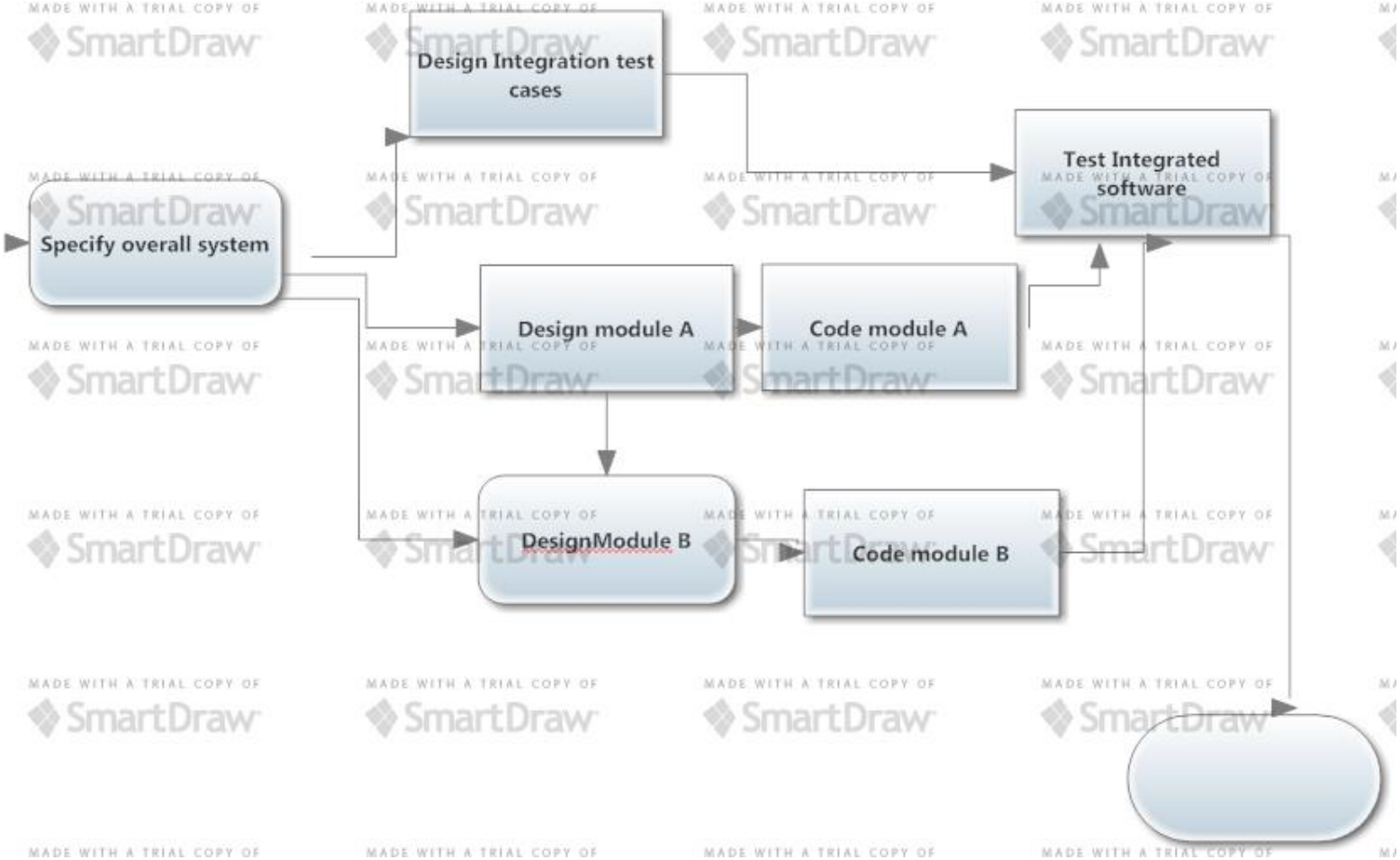
For example, in module products there may be two software modules to be built. For each module, design and code development are to be performed.

4.4 Product ideal activity network: Activity network shows the tasks that have to be carried out and the order in which they have to be examined

For product creation, identification of activities and their order of execution is to be decided.

Example of an activity network is shown in the next slide:

Step 4 contd..


Design Integration test cases
Test Integrated software
Specify overall system
Design module A
Code module A
DesignModule B
Code module B

Stepwise Project Planning contd..

4.5 Modify the ideal activity network to include stages and checkpoints: Ideal in the sense that resource constraints are not considered

Checkpoints are required to monitor important stages of a project. The stages indicate completion of important activities.

5. Estimate Effort for Each Activity

5.1 Carry out bottom-up estimates: For each activity the time, the (Staff) effort required are to be identified.

They vary depending on activity. All these times and efforts summed should be overall time and effort required for the project.

5.2: Revise Plan to create controllable activities: As it is difficult to control long activities with long duration, such long activities need to be divided into smaller sub-tasks.

Stepwise Project Planning contd...

Step 6: Identify Activity Risks

6.1: Identify and quantify activity based risks: The possibility that an assumption upon which a plan is based is incorrect constitutes a risk

For example, the design of a component is planned to take 5 days based on the assumption that the client's requirement is clear

If it is not clear additional effort to clarify the requirement could be needed. Hence, the estimate of effort can be expressed as a range of values.

Step 6 Contd..

6.2 Plan risk reduction and contingency measures where appropriate: If possible measures are to be taken to avoid or reduce some of the identified risks

Contingency plan is an action to be taken if a risk occurs. For example to use contract staff if a project member of a team is unavailable at a key time.

6.3 Adjust overall plans and estimates to take account of risks: Change the plans by adding new activities which reduce risks

For example a new programming language is to be used for coding, training and time for programmers to practice are to be additionally provided

Stepwise Project Planning contd...

Step 7: Allocate Resources

7.1: Identify and Allocate Resources: The available staff are allocated to tasks. The staff needed for each activity is identified.

7.2: Revise plans and estimates to take into account resource constraints: Generally there will be constraints on resources

For example some staff may be involved at the same time in more than one task; so order of priority is required

This can effect the overall project duration. Hence plans and estimates are to be revised based on constraints.

Stepwise Project Planning contd...

Step 8: Review or Publicize Plans

8.1 Review quality aspects of the project plan: Each task should have quality criteria. Quality checks have to be passed before the activity can be signaled as completed

8.2 Document plans and obtain agreement: The plans need to be carefully documented

All the stakeholders of the project understand and agree to the commitments required of them

Stepwise Project Planning contd...

Steps 9 and 10: Execute plan/Lower level of planning

Once the plan is executed for the project, detailed plans for each activity are needed

Provisional plans for the more distant tasks or the tasks for later stages of the project can be made

Conclusion

- In this stepwise project planning a number of activities are included

  Planning is an iterative process

  while executing activity, detailed replanning is needed.

- Any planning approach should have the following
  - Project objectives
  - Analysis of project characteristics such as it is product driven, or objective driven, taking into account project risks, considering user requirements, and planning resource estimates

Conclusion contd..

- Establishment of an infrastructure with a set of standards methods and tools.
- Identification of sub products and activities needed to generate these.
- Allocation of resources to activities
- Establishment of quality controls
